

# G. Leigh Harwell

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## Education:

University of Oklahoma — Norman, OK

**Bachelor of Science in Astrophysics** | Minor in Mathematics

Expected Graduation: May 2027 | GPA: 3.70

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## Research Experience:

### Multi-Messenger Observations of Compact Objects

Department of Physics and Astronomy, University of Oklahoma

Undergraduate Capstone Research | August 2026 - present

- Characterizing double white dwarf systems through multi messenger observations.
- Analyze multi messenger datasets to identify compact binary candidates for follow-up observational studies.

### Machine Learning and Galactic Archaeology

Department of Physics and Astronomy, University of Oklahoma

Undergraduate research | October 2024 - October 2026]

- Apply machine learning techniques to asteroseismology and galactic archaeology datasets to improve stellar classification models.

### NSF Research Experience for Undergraduates (REU)

Department of Physics and Astronomy, University of Oklahoma | June 2026 – July 2026

- Implementing “TheCannon” data driven model to estimate stellar parameters in Astroseismolog with elemental abundances from spectroscopic data.
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## Awards and Honors:

**UReCA Fellowship** | The University of Oklahoma | \$5,000 Grant | 2026

- Funded collaborative research developing machine learning tools for stellar label estimation; presented findings at department symposium.

**Homer L. Dodge Department of Physics and Astronomy Meritorious Scholarship** | 2024 - present

- Awardees to outstanding physics and astronomy students that maintain a 4.0 GPA throughout the semester

**President's Honor Roll** | The University of Oklahoma

- award to distinguished students that maintain a 4.0 GPA throughout the semester
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## **Skills:**

**Programming Languages** | Python and LaTeX, experience in scientific libraries and data analysis tools and models. Scientific Libraries: NumPy, SciPy, Pandas, Matplotlib, Astropy, scikit-learn

**Data Analysis** | Multi-messenger data analysis, Stellar parameter modeling, Machine Learning workflows

**Problem Solving** | utilize a variety of skills and methods to find creative solutions to a wide array of problems

**Computer Literacy** | utilize various tools in the operation of computers and various other technology

**Adaptability** | respond flexibly to evolving factors, environments, and conditions by using new skills

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## **Teaching and Leadership Experience:**

**Physics for Life Science Majors Tutoring Coordinator & PLA** | August 2025 — Present

Action Tutoring | Student Learning Center | University of Oklahoma, Wagner Hall, Norman, OK

- Coordinating the tutoring for the life science majors, and collaborating with department faculty to provide extra problem sets and conceptual worksheets for exam review.
  - Provide individual and small-group tutoring sessions in introductory mechanics, optics, thermodynamics, and electromagnetism.
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## **Extracurriculars:**

**Community and Belonging** | August 2026 — Present

**Society of Physics Students** | August 2023 — Present

**Women in Physics** | August 2023 — Present